

QnA - Customer Shopping Behaviour 2 (SQL)

1. Connect to the existing SQLite database

```
In [110... import sqlite3
import pandas as pd

conn = sqlite3.connect('shopping_behavior.db')

print("Successfully connected to 'shopping_behavior.db'!\n")
```

Successfully connected to 'shopping_behavior.db'!

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Q1. Total revenue generated by male vs. female customers

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```
In [111... q1_result = pd.read_sql("""
SELECT
    Gender,
    SUM("Purchase Amount (USD)") AS Total_Revenue
FROM customer_transactions

GROUP BY Gender;

""", conn)

q1_result
```

```
Out[111...
   Gender  Total_Revenue
0  Female           75191
1   Male          157890
```

Q2. Customers who used a discount but spent more than the average purchase amount

In [112...

```
q2_result = pd.read_sql("""  
  
SELECT "Customer ID", Age, Gender, Category, "Purchase Amount (USD)", "Discount Ap  
FROM customer_transactions  
  
WHERE "Discount Applied" = 'Yes' AND "Purchase Amount (USD)" > (SELECT AVG("Purchas  
FROM customer_transactions);  
  
""", conn)  
  
print(f"\nTotal customers who used a discount but spent more than the average purch  
df = q2_result  
q2_result
```

Total customers who used a discount but spent more than the average purchase amount are: 839

Out[112...

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Discount Applied
0	2	19	Male	Clothing	64	Yes
1	3	50	Male	Clothing	73	Yes
2	4	21	Male	Footwear	90	Yes
3	7	63	Male	Clothing	85	Yes
4	9	26	Male	Outerwear	97	Yes
...	...	...	...	...	...	...
834	1667	51	Male	Clothing	64	Yes
835	1671	22	Male	Clothing	73	Yes
836	1673	18	Male	Footwear	73	Yes
837	1674	21	Male	Clothing	62	Yes
838	1676	35	Male	Clothing	90	Yes

839 rows × 6 columns

In [113...

```
# Save the data dataframe to a new CSV file
df.to_csv('SQL-customer_applied_dis_aboveAVG.csv', index=False)
```

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Q3. Top 5 products with the highest average review rating

=====



In [114...

```
q3_result = pd.read_sql("""
SELECT
    "Item Purchased",
    ROUND(AVG("Review Rating"), 2) AS Avg_Review_Rating
FROM customer_transactions

GROUP BY "Item Purchased"
ORDER BY Avg_Review_Rating DESC

LIMIT 5;

""", conn)
```

q3_result

Out[114...

	Item Purchased	Avg_Review_Rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Skirt	3.79

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Q4. Compare average purchase amounts between Standard and Express Shipping

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Average purchase amounts of - All shipping types

In [115...

```
q4_result = pd.read_sql("""
SELECT
    "Shipping Type",
    ROUND(AVG("Purchase Amount (USD)"), 2) AS Avg_Purchase_Amount
FROM customer_transactions
GROUP BY "Shipping Type";
""", conn)

q4_result
```

Out[115...

	Shipping Type	Avg_Purchase_Amount
0	2-Day Shipping	60.73
1	Express	60.48
2	Free Shipping	60.41
3	Next Day Air	58.63
4	Standard	58.46
5	Store Pickup	59.89

Average purchase amounts between Standard and Express Shipping

```
In [116...] q4_result = pd.read_sql("""
SELECT
    "Shipping Type",
    ROUND(AVG("Purchase Amount (USD)"), 2) AS Avg_Purchase_Amount
FROM customer_transactions

WHERE "Shipping Type" IN ('Standard', 'Express')
GROUP BY "Shipping Type";

""", conn)

q4_result
```

```
Out[116...]
  Shipping Type  Avg_Purchase_Amount
0      Express              60.48
1      Standard              58.46
```

=====

Q5. Compare average spend and total revenue between subscribers and non-subscribers

=====



```
In [117...] q5_result = pd.read_sql("""
SELECT
    "Subscription Status",
    COUNT("Customer ID") AS Total_Customers,
    ROUND(AVG("Purchase Amount (USD)"), 2) AS Avg_Spend,
    SUM("Purchase Amount (USD)") AS Total_Revenue
FROM customer_transactions

GROUP BY "Subscription Status";

""", conn)
```

q5_result

Out[117...

	Subscription Status	Total_Customers	Avg_Spend	Total_Revenue
0	No	2847	59.87	170436
1	Yes	1053	59.49	62645

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Q6. Top 5 products with the highest percentage of purchases with discounts applied

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In [118...

```
q6_result = pd.read_sql("""
SELECT
    "Item Purchased",
    ROUND(100.0 * SUM(CASE WHEN "Discount Applied" = 'Yes' THEN 1 ELSE 0 END) /
FROM customer_transactions

GROUP BY "Item Purchased"
ORDER BY "Highest percentage of purchases with discounts" DESC

LIMIT 5;
""", conn)

q6_result
```

Out[118...

	Item Purchased	Highest percentage of purchases with discounts
0	Hat	50.00
1	Sneakers	49.66
2	Coat	49.07
3	Sweater	48.17
4	Pants	47.37

Seeing which products rely heavily on the discount.

Q7. Segment customers into New, Returning, and Loyal based on previous purchases

In [119...

```
q7_result = pd.read_sql("""  
  
    SELECT  
        CASE  
            WHEN "Previous Purchases" == 1 THEN 'New'  
            WHEN "Previous Purchases" BETWEEN 2 AND 10 THEN 'Returning'  
            ELSE 'Loyal'  
        END AS Customer_Segment,  
  
        COUNT("Customer ID") AS Segment_Count  
  
    FROM customer_transactions  
    GROUP BY Customer_Segment;  
  
""", conn)  
  
q7_result
```

Out[119...

	Customer_Segment	Segment_Count
0	Loyal	3116
1	New	83
2	Returning	701

There are lots of loyal customers in this data set.

END: Marks the finish line for the CASE statement

Customer_Segment so you can easily reference it in your GROUP BY

Q8. Top 3 most purchased products within each category

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```
In [120... q8_result = pd.read_sql("""
    WITH RankedItems AS (
        SELECT Category, "Item Purchased", COUNT(*) AS Purchase_Count,
            ROW_NUMBER() OVER(PARTITION BY Category ORDER BY COUNT(*) DESC) as rn
        FROM customer_transactions
        GROUP BY Category, "Item Purchased"
    )
    SELECT Category, "Item Purchased", Purchase_Count
    FROM RankedItems
    WHERE rn <= 3;
""", conn)

q8_result
```

Out[120...

	Category	Item Purchased	Purchase_Count
0	Accessories	Jewelry	171
1	Accessories	Sunglasses	161
2	Accessories	Belt	161
3	Clothing	Pants	171
4	Clothing	Blouse	171
5	Clothing	Shirt	169
6	Footwear	Sandals	160
7	Footwear	Shoes	150
8	Footwear	Sneakers	145
9	Outerwear	Jacket	163
10	Outerwear	Coat	161

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Q9. Are repeat buyers (more than 5 previous purchases) also likely to subscribe?

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```
In [121... q9_result = pd.read_sql("""
SELECT
    CASE WHEN "Previous Purchases" > 5 THEN 'Repeat Buyer (>5)' ELSE 'Low Frequ
    "Subscription Status",
    COUNT("Customer ID") AS Customer_Count
FROM customer_transactions
GROUP BY Buyer_Type, "Subscription Status";
""", conn)

q9_result
```

Out[121...

	Buyer_Type	Subscription Status	Customer_Count
0	Low Frequency (<=5)	No	329
1	Low Frequency (<=5)	Yes	95
2	Repeat Buyer (>5)	No	2518
3	Repeat Buyer (>5)	Yes	958

Finding: Yes, repeat buyers show a slightly higher tendency to subscribe compared to low-frequency buyers, though overall subscription rates remain moderate across both groups.

Data Breakdown:

Repeat Buyers (>5 Previous Purchases): Out of 3,476 repeat customers, 958 are active subscribers, giving a subscription rate of ~27.56%.

Low-Frequency Buyers (≤5 Previous Purchases): Out of 424 customers, 95 are active subscribers, yielding a subscription rate of ~22.41%.

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Q10. Revenue contribution of each age group

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```
In [122... q10_result = pd.read_sql("""
```

```

SELECT
    "Age Group",
    SUM("Purchase Amount (USD)") AS Total_Revenue,
    ROUND(AVG("Purchase Amount (USD)"), 2) AS Avg_Spend
FROM customer_transactions

GROUP BY "Age Group"
ORDER BY Total_Revenue DESC;

""" , conn)

q10_result

```

Out[122...

	Age Group	Total_Revenue	Avg_Spend
0	Young Adult	62143	60.45
1	Middle-aged	59197	60.04
2	Adult	55978	59.42
3	Senior	55763	59.07

In [123...

```

# Close the database connection when done
conn.close()
print("Database connection closed successfully.")

```

Database connection closed successfully.